Tutorial 3: Quasielastic Neutron Scattering (QENS)

Chi Y. Cheng, Maciej Bartkowiak, Sanghamitra Mukhopadhyay

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Tutorial 3: Quasielastic Neutron Scattering (QENS)

- Elastically scattered neutron have the same energy before and after scattering. This leads to a sharp peak at ω=0.
- Quasielastic scattering is a special case of inelastic scattering and leads to a broad peak around ω=0. QENS that can be used to determine diffusion constant of liquids and solids.
- We'll be looking at a paper by Sköld et al.

PHYSICAL REVIEW A VOLUME 6, NUMBER 3 SEPTEMBER 1972

Coherent- and Incoherent-Scattering Laws of Liquid Argon†

K. Sköld,* J. M. Rowe, and G. Ostrowski
Solid State Science Division, Argonne National Laboratory, Argonne, Illinois 60439
and
P. D. Randolph‡
Nuclear Technology Division, Idaho Nuclear Corporation, Idaho Falls, Idaho
(Received 17 January 1972)

Phys. Rev. A **6**, 1107 – **1972**
DOI: https://doi.org/10.1103/PhysRevA.6.1107

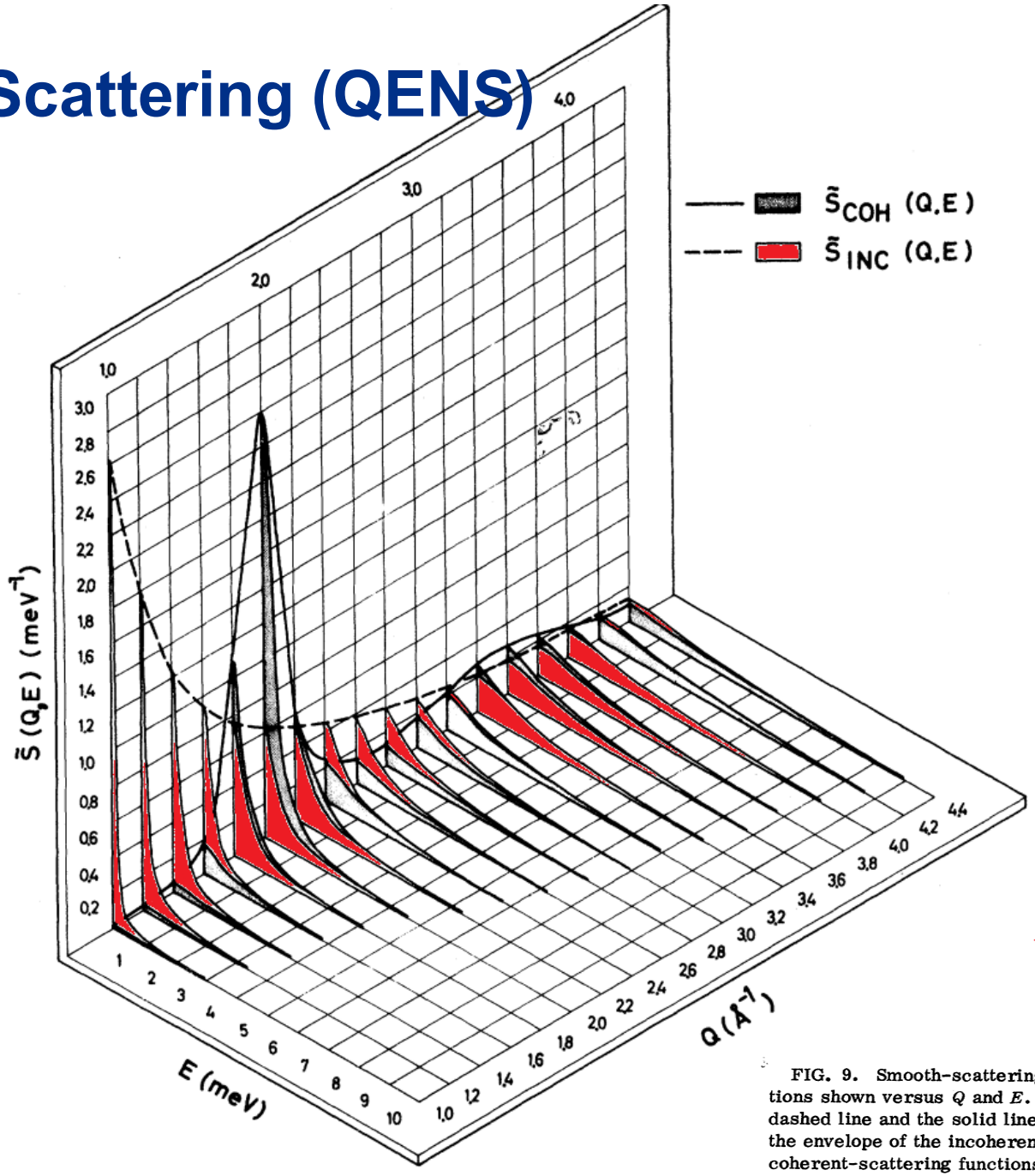


FIG. 9. Smooth-scattering functions shown versus Q and E. The dashed line and the solid line show the envelope of the incoherent and coherent-scattering functions, respectively.

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- In this paper, neutrons were obtained from a reactor source (Materials Testing Reactor - Idaho National Laboratory – decommissioned 2011).

Experimental setup:

- Neutron instrument with a velocity selector (chopper-based)
- Time of flight inelastic instrument using thermal neutrons

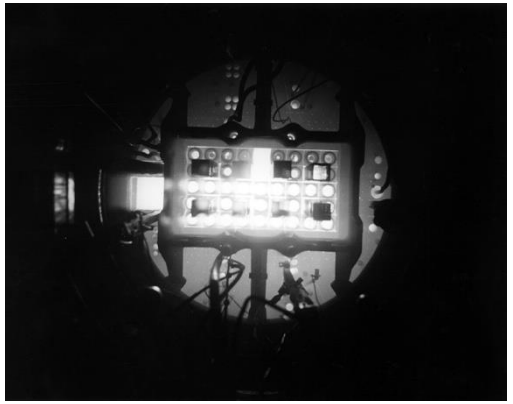


Image: View down through 20 feet of water into the tank of Materials Testing Reactor 1955

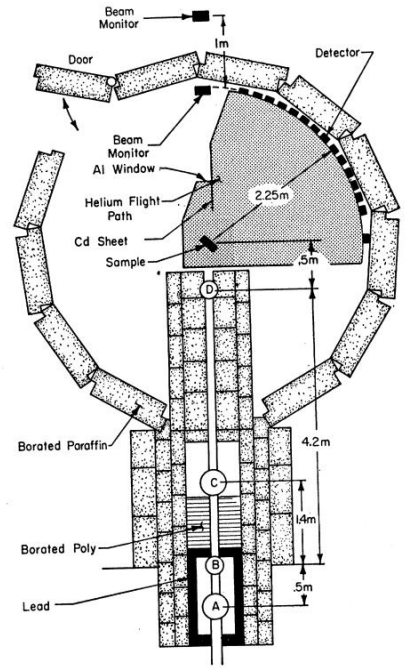


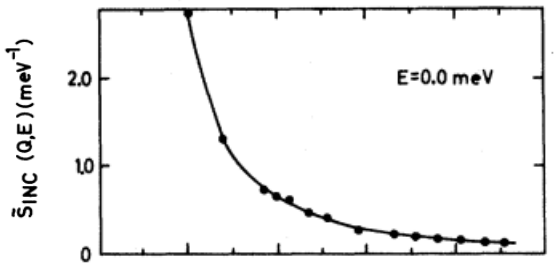
FIG. 2. Plan view of the velocity selector at the HT-1 through-hole showing the chopper arrangements, pertinent distances, and the helium-filled flight path.

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- The QENS peak is a Lorentz function.
- By measuring the half-width at half-maximum of the QENS peak for a number of values of q the diffusion constant (D) of the system can be determined.

$$S_{\mathrm{i}}(q,\omega) = \frac{1}{\pi}\frac{\hbar\Gamma(q)}{(\hbar\omega)^2 + (\hbar\Gamma(q))^2}$$

$$\Gamma(q) = Dq^2$$



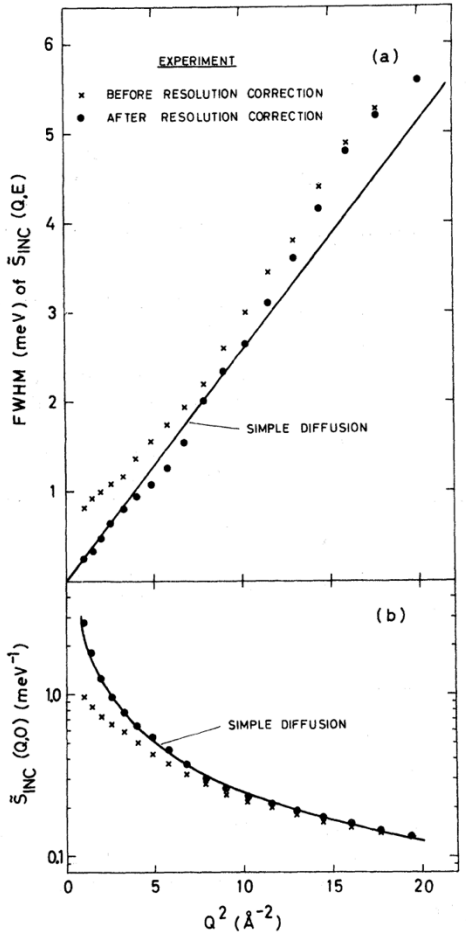


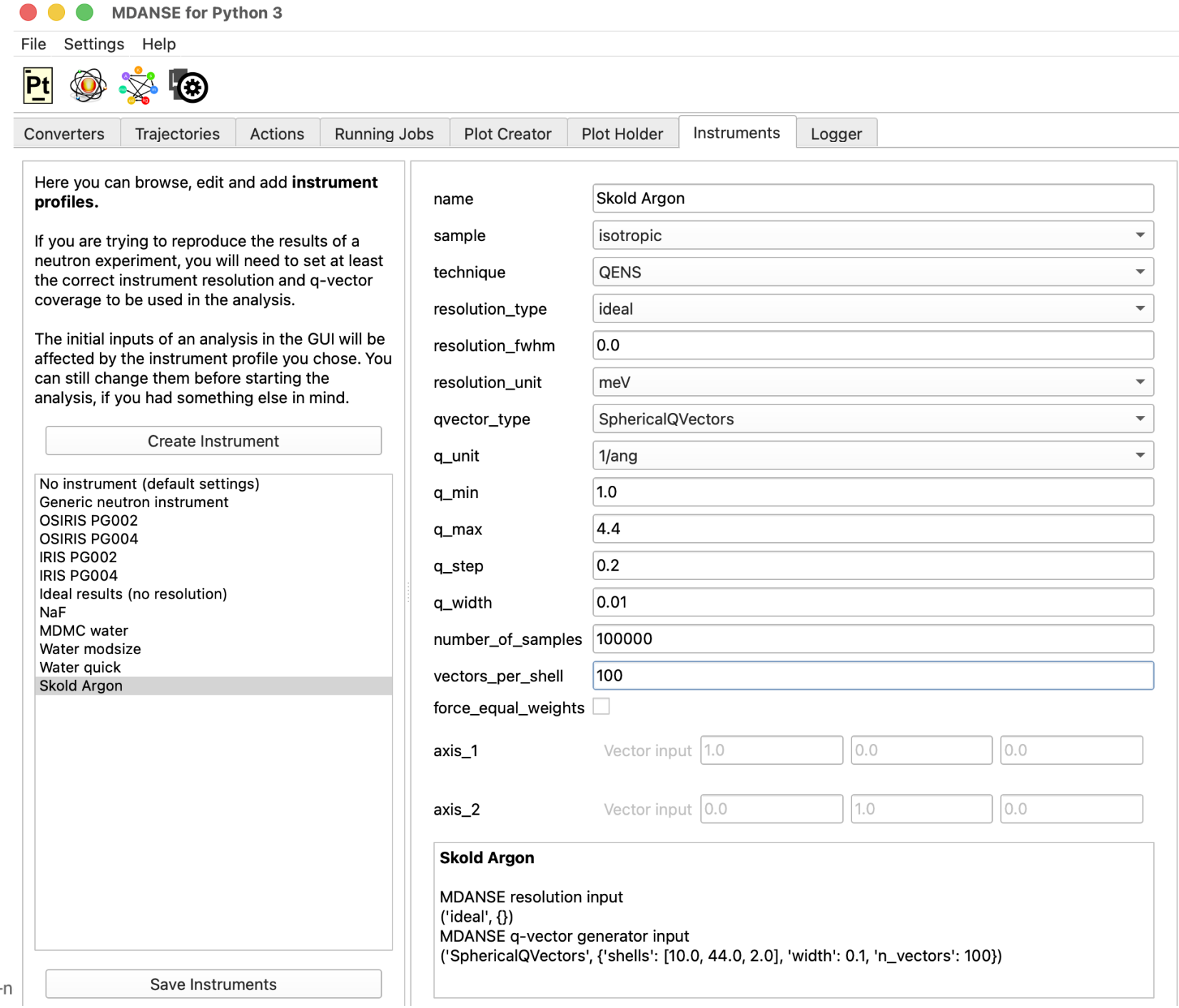
FIG. 11. (a) Full width at half-maximum of the incoherent-scattering function before and after the resolution correction. Also shown is the simple diffusion result with $D = 1.94 \times 10^{-5}$ cm^2/sec (Ref. 48). (b) Intensity at zero-energy transfer for the incoherent-scattering function before and after the resolution correction. The solid line shows the simple diffusion result.

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- Let's calculate the QENS peaks and determine the diffusion constant from it following the paper from Sköld et at.
- Load up the Argon MDT trajectory `ar36_120fs_8_f64.mdt`.
- First, we will save the scattering parameters as an instrument profile.
- Then, we will calculate the Dynamic Incoherent Structure Factor (DISF) and determine the diffusion constant by fitting the results.

Tutorial 3: Save your scattering parameters

- Go to the Instruments tab
- Click "`Create Instrument`"
- Give it a meaningful name
- Pick `resolution_type: ideal`
- Pick `qvector_type: SphericalQVectors`
- Set q values from 1.0 to 4.4 in steps of 0.2 $Å^{-1}$
- `vectors_per_shell: 100`
- `width: 0.01`
- `Save Instruments`



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Go to the Actions tab.

Set the instrument to the one you just created.

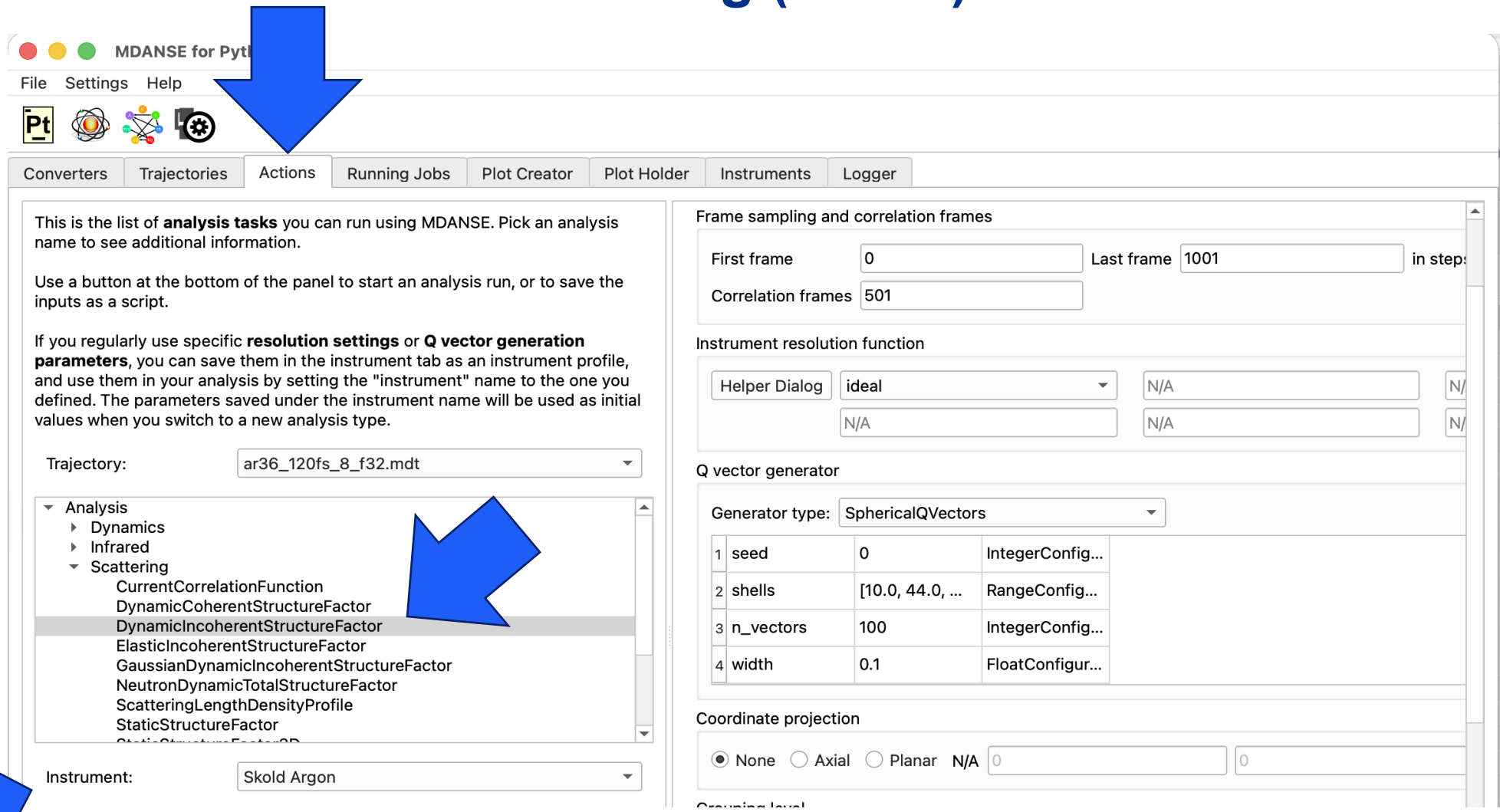
Choose:

Analysis ->

Scattering ->

DynamicIncoherent

StructureFactor



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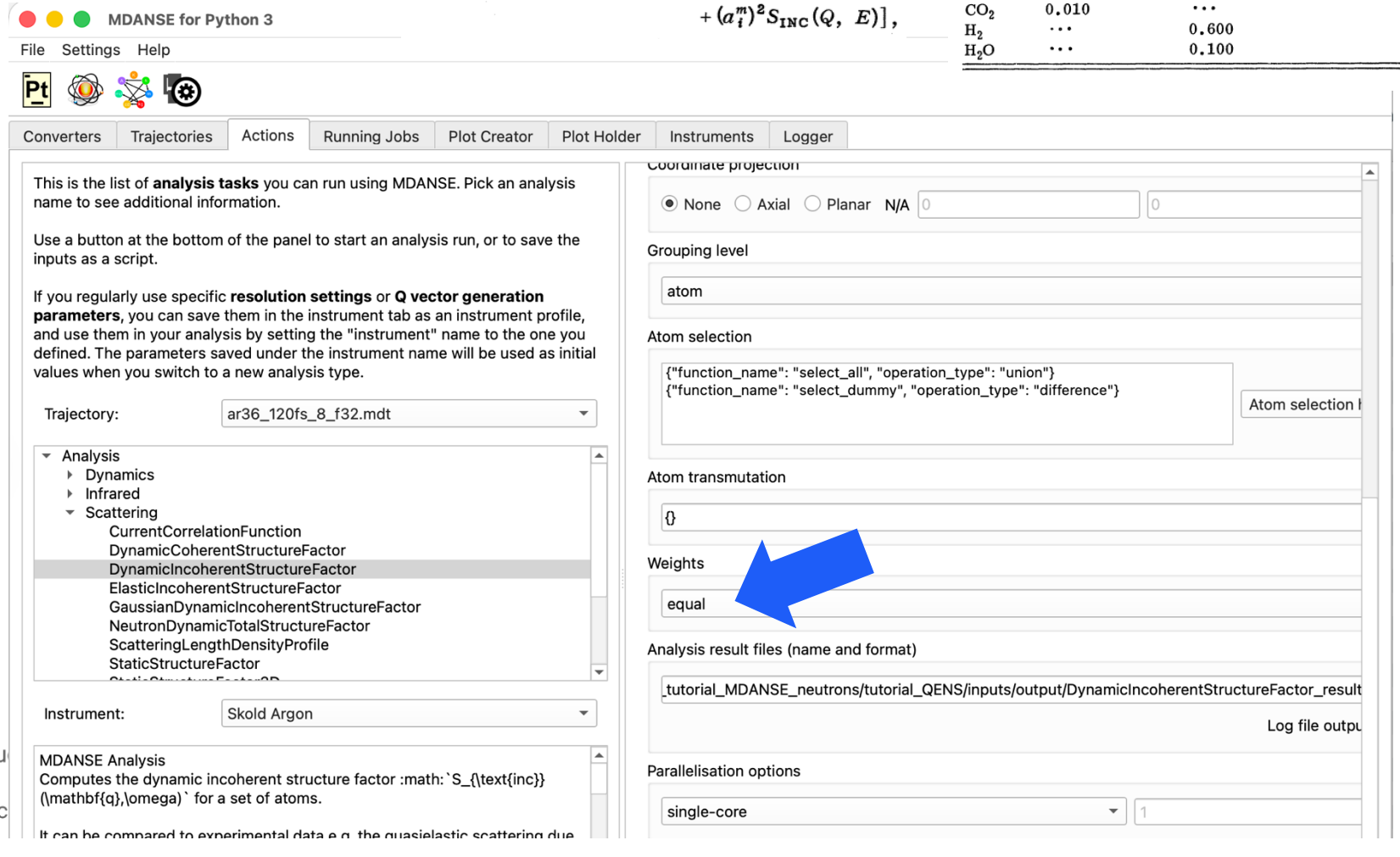
- Set the weight to equal.
- Our trajectory is pure Ar36 which does not scatter incoherently – we set equal to get an unweighted S_inc(q,f).
- Experimentally Sköld et al. obtained S_inc(q,f) by running scattering experiments on two different Argon samples and solving for S_inc(q,f).

$$S^{36}(Q,\ E) = \frac{1}{(a_c^{36})^2 + (a_i^{36})^2}\left[(a_c^{36})^2 S_{\mathrm{COH}}(Q,\ E) + (a_i^{36})^2 S_{\mathrm{INC}}(Q,\ E)\right],$$

$$S^{m}(Q,\ E) = \frac{1}{(a_c^{m})^2 + (a_i^{m})^2}\left[(a_c^{m})^2 S_{\mathrm{COH}}(Q,\ E) + (a_i^{m})^2 S_{\mathrm{INC}}(Q,\ E)\right],$$

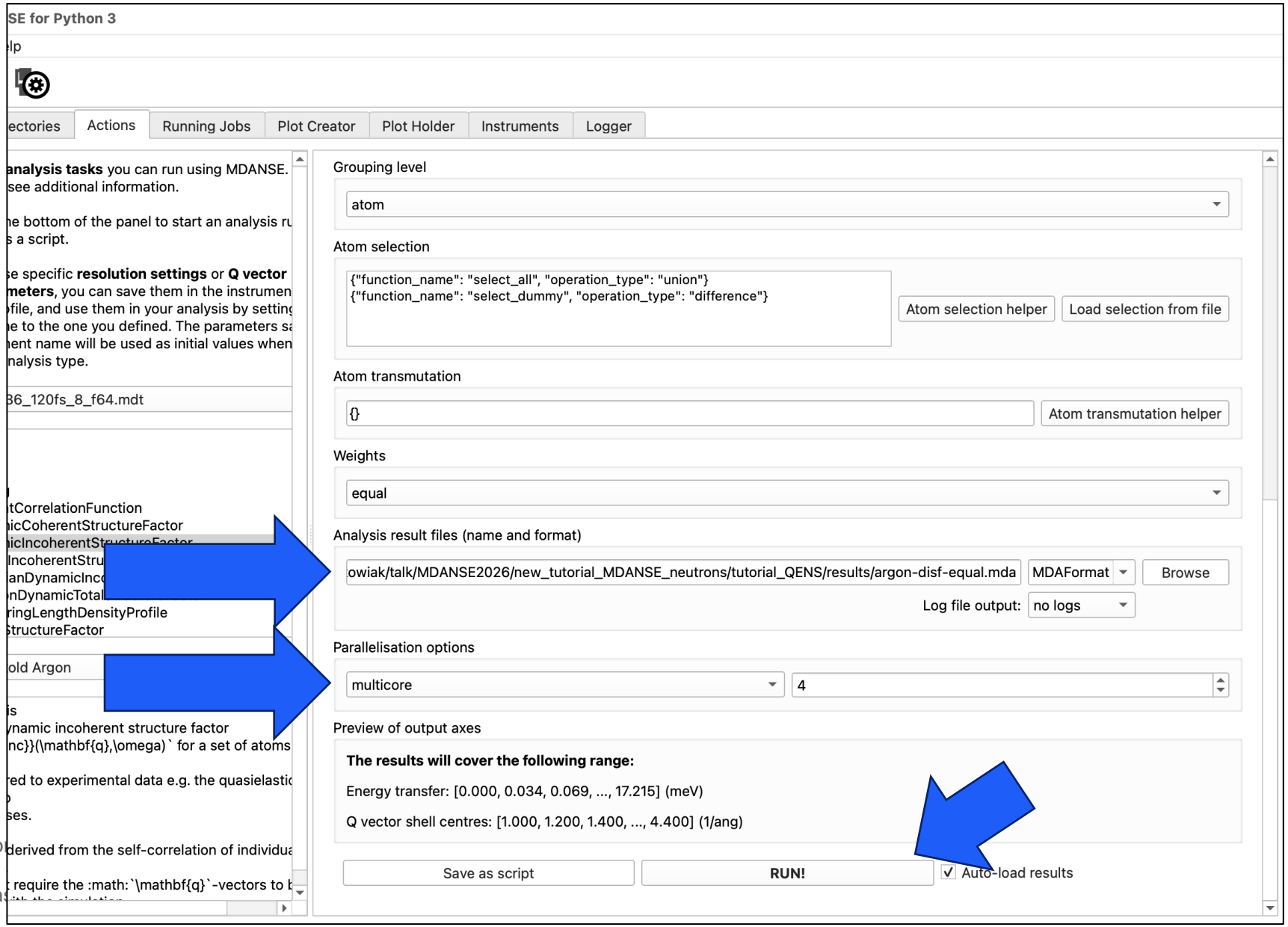
TABLE I. Composition of samples and scattering cross sections for argon isotopes.

	A^{36} sample	Mixture sample	σ_s (b)
A^{36}	98.535	7.128	77.8 ± 0.4
A^{38}	0.815	0.099	1.6 ± 1.6
A^{40}	0.080	91.773	0.419 ± 0.003
N_2	0.550	0.300	
O_2	0.010	...	
CO_2	0.010	...	
H_2	...	0.600	
H_2O	...	0.100	



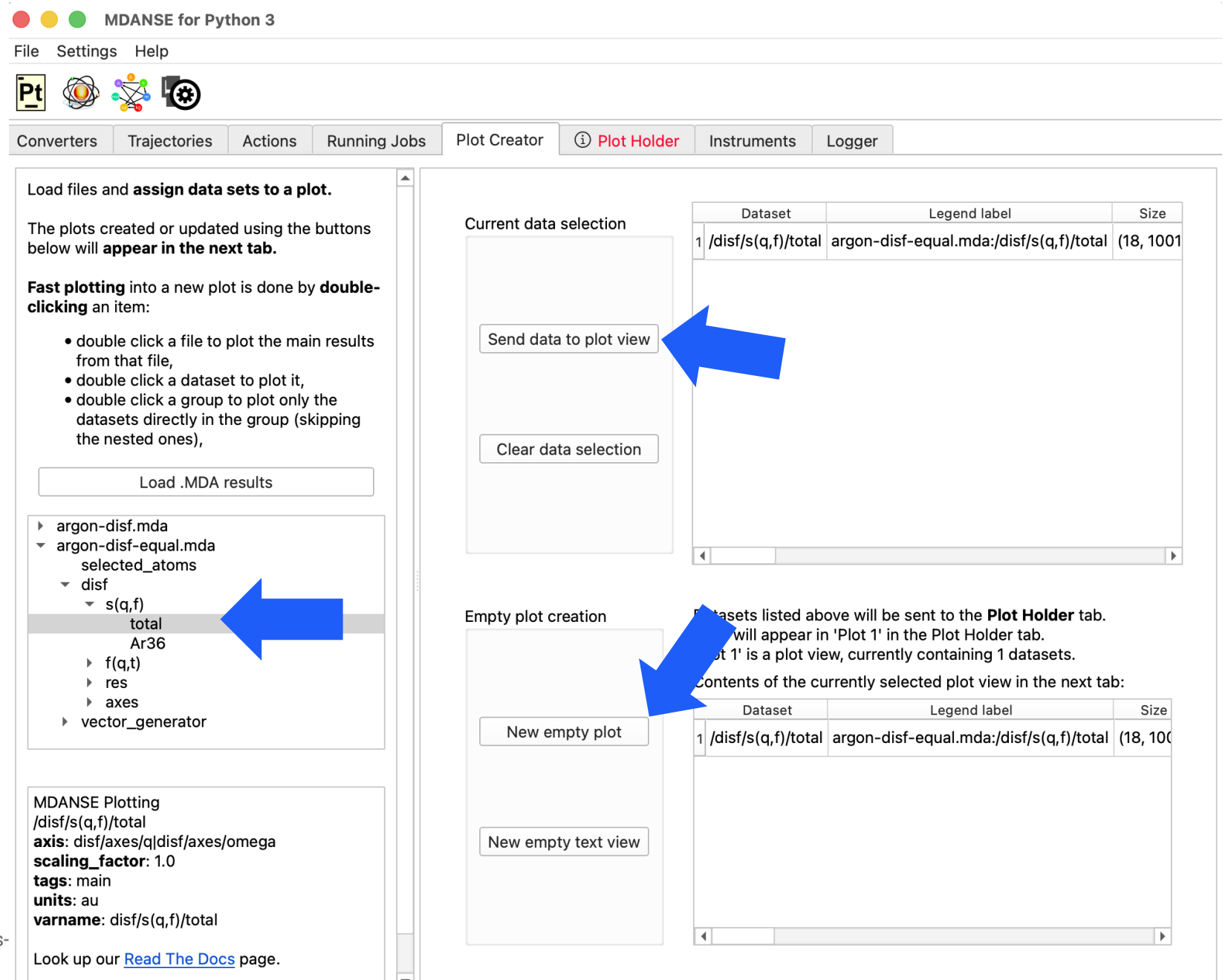
Tutorial 3: Quasielastic Neutron Scattering (QENS)

- Set the output file, set multicore with 2 or more cores and hit “RUN!”
- Check the “Running Jobs” tab - this should take less than 3 minutes.



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- Once complete, go to the Plot Creator and create a plot of `S(q,f)_total`.



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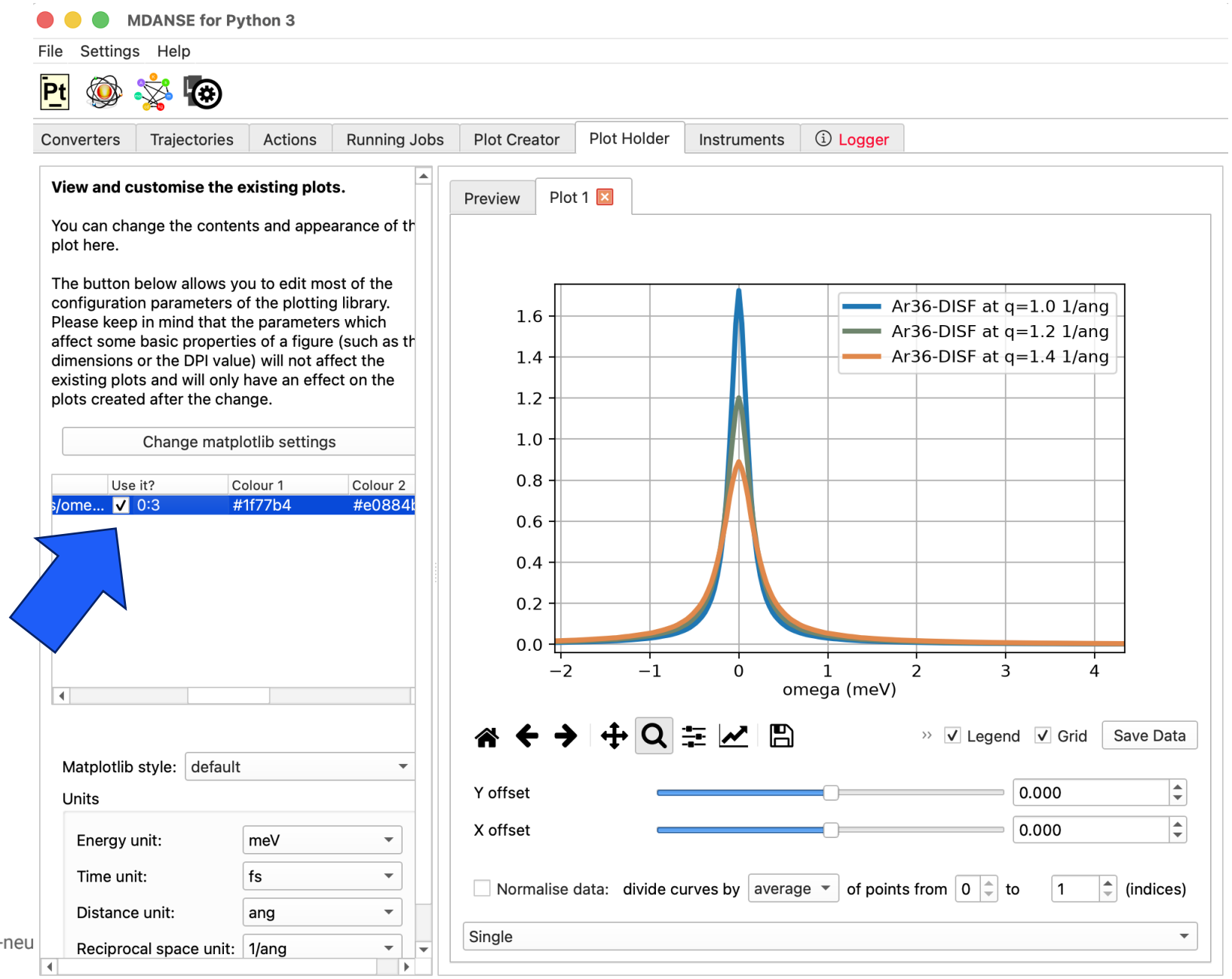
- Using your cursor determine the half width at half maximum for each plot.
- The first plot is |Q|=1.0 Å^{-1}, the next is |Q|=1.2 Å^{-1}, and etc.
- You can enter a number:

`4`

or a range:

`0:3`

in the "Use it?" column to select curves corresponding to specific `q_vector` magnitudes.



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- Using a spreadsheet to create a plot to determine the diffusions constant.
- HINT: see FIG. 11 in the paper by Sköld et al.

$$S_{\mathrm{i}}(q,\omega) = \frac{1}{\pi}\frac{\hbar\Gamma(q)}{(\hbar\omega)^2 + (\hbar\Gamma(q))^2}$$

$$\Gamma(q) = Dq^2$$

Q (ang-1)	Q (cm-1)	Q^2 (cm-2)	HWHM (meV)	HWHM / hbar (s-1)
1				
1.2				
1.4				
1.6				
1.8				
2				
2.2				
2.4				
2.6				
2.8				
3				
3.2				
3.4				
3.6				
3.8				
4				
4.2				
4.4				

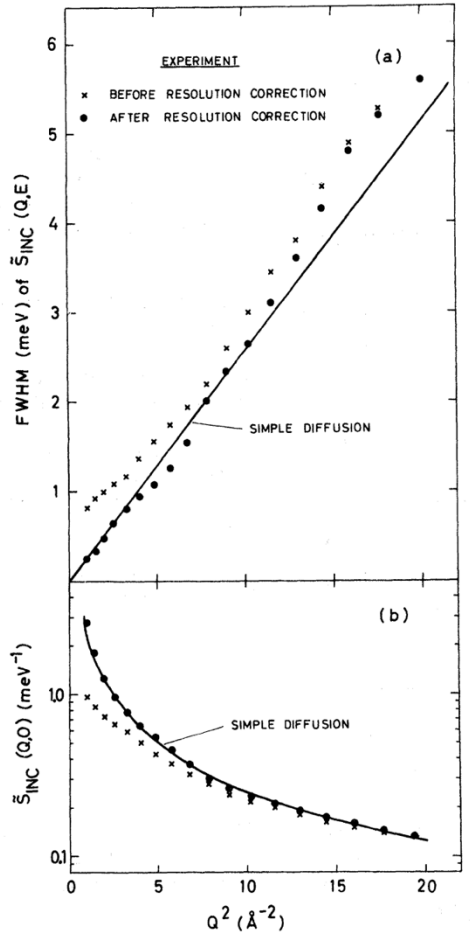
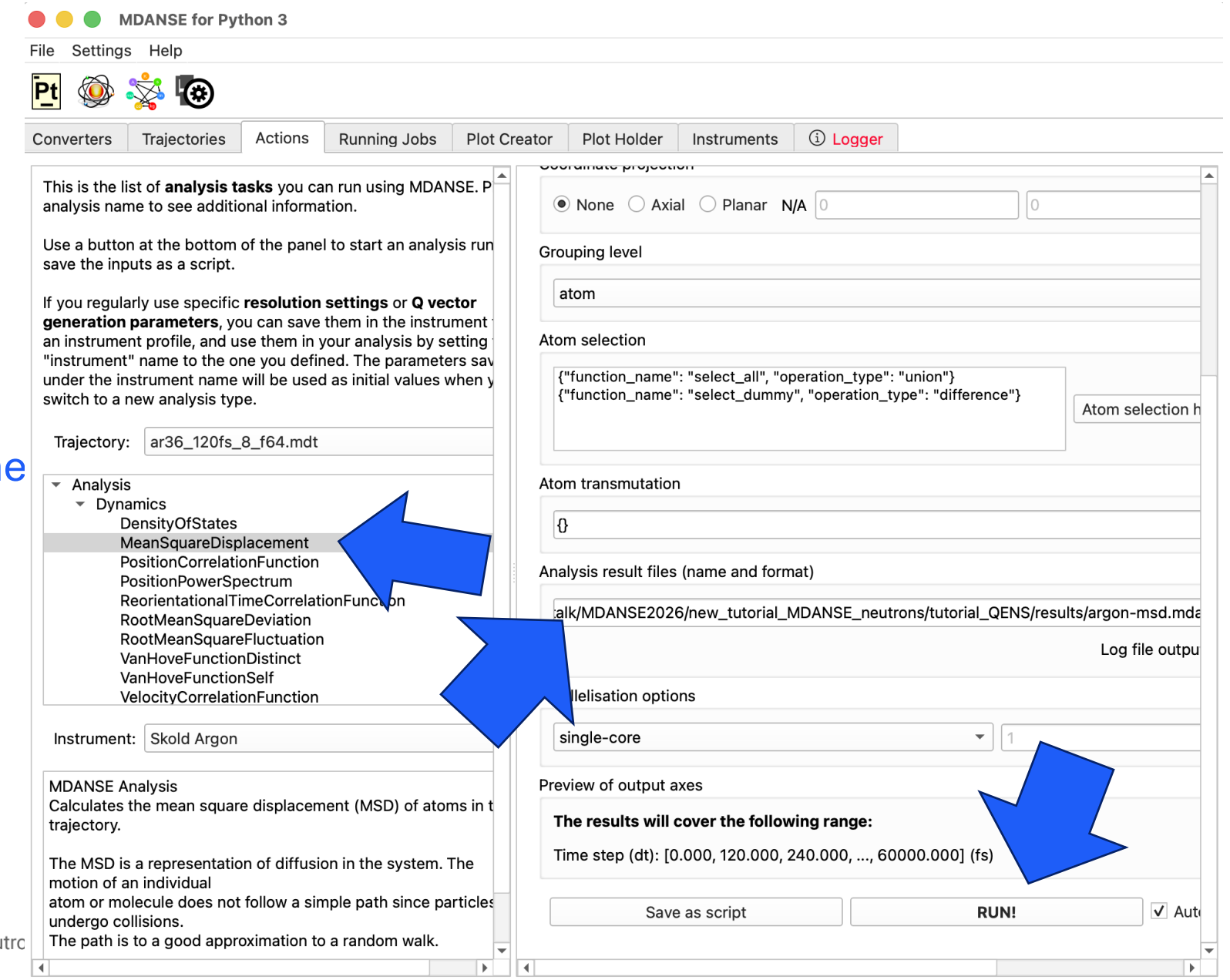


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- Now let's calculate the diffusion constant directly from the mean square displacement and compare results.
- Using the `ar36_120fs_8_f64.mdt` run the mean square displacement calculation.
- Default settings are fine. Just set the output path and run it



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- The Argon MSD is essentially linear for long times, we can calculate the diffusion constant with.

$$D = \frac{\mathrm{MSD}(t)}{6t}$$

- Calculate the diffusion constant using the last time step t=60000 fs.
- Did your results from QENS and MSD agree with the experimental?

